

RemoteZero: Geospatial Reasoning with Zero Labels

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Geospatial reasoning requires models to identify image regions that satisfy complex and often implicit user intents. Recent reinforcement learning approaches improve reasoning without manually annotated reasoning traces, but still require human-provided target labels to construct rewards, limiting their use on large-scale unlabeled Earth observation data. We introduce RemoteZero, a label-free framework for reinforcement-based geospatial reasoning. Our key observation is twofold: MLLMs are often more reliable at evaluating candidates than generating solutions, while aerial imagery reduces interference in region-level verification. Therefore, RemoteZero converts each predicted region into a visual crop and uses its semantic consistency with the query as an intrinsic reward for GRPO optimization. This formulation removes the need for human-provided solution labels and further supports iterative self-evolution by reusing previous-round models as verifiers. Experiments show that RemoteZero outperforms strong supervised baselines, applies effectively to other Earth observation tasks, and continues to improve through self-evolution as the training data expand. We hope this direction can broadly benefit the Earth observation community.

Keywords: Remote sensing, Geospatial reasoning, Multimodal large language models, Label-free, Reinforcement learning

 **Code Repository:** <https://github.com/1e12Leon/RemoteZero>

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1. Introduction

The long-term objective of remote sensing foundation models Xu et al. (2025), Zhang et al. (2024a), Zhou et al. (2024b) is to advance Earth science, in which many real-world tasks carry substantial social and economic value. Geospatial reasoning Li et al. (2025), Yao et al. (2026a) is a key task toward this objective, as it requires models to identify regions that satisfy implicit, non-technical queries rather than predefined object categories. For example, after an earthquake, a decision-maker is unlikely to ask merely to “Detect a playground.” A realistic request may instead be to “Identify an optimal zone for resettling victims that maximizes capacity while facilitating rapid facility deployment.” Such a query does not name a specific visual target; it describes a functional need whose solution depends on jointly interpreting the intended use, spatial constraints, and scene evidence. The model must infer which visible region best serves the intended function, shifting remote sensing from content recognition toward decision-oriented reasoning.

Recent RFT/GRPO-based studies have advanced tasks adjacent to geospatial reasoning, including referring expression segmentation Fiaz et al. (2025), Zhang et al. (2026) and visual grounding Wang et al. (2025a), Li et al. (2026b), Sun et al. (2026). However, these tasks typically specify the target explicitly, requiring models to identify a given object or referent rather than infer the latent functional intent behind a decision-oriented query. SegEarth-R1 Li et al. (2025) pioneered the Geospatial Pixel

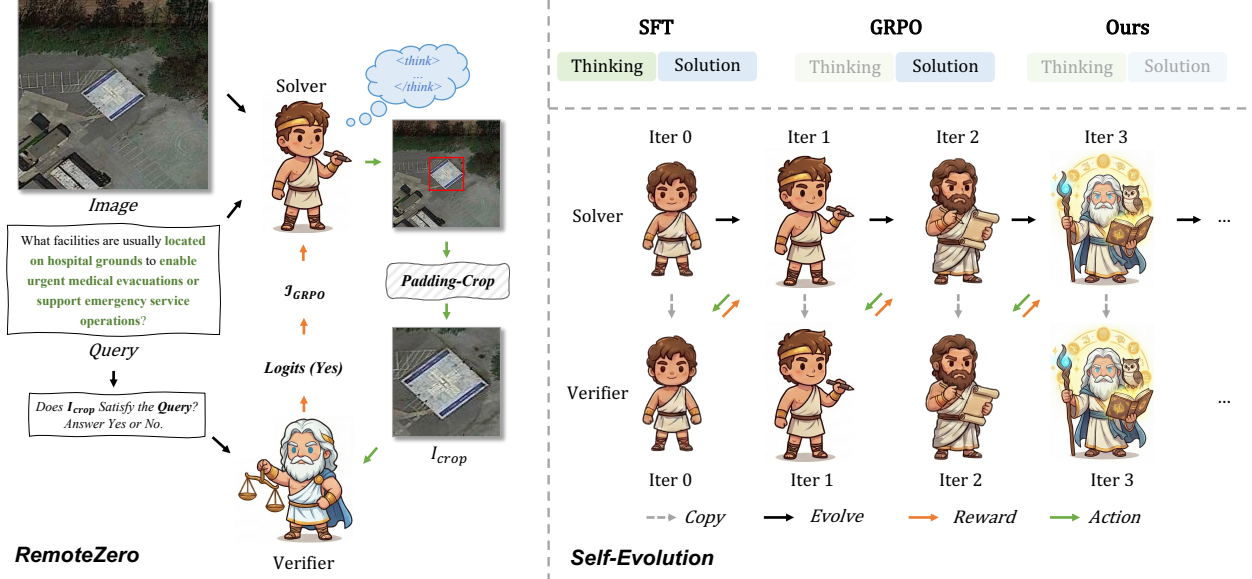


Figure 1: Conceptual overview of RemoteZero. Left: predicted regions are cropped and semantically verified to produce label-free rewards. Top-right: comparison with SFT and conventional GRPO. Bottom-right: previous-round models are reused as verifiers for iterative self-evolution.

Reasoning task, employing Supervised Fine-Tuning (SFT) Wang et al. (2025b), Han et al. (2024) to align models with manually curated triplets of images, reasoning chains, and segmentation masks. However, this paradigm depends on labor-intensive annotations and may overfit to fixed reasoning patterns. To reduce the reliance on annotated reasoning traces, RemoteReasoner Yao et al. (2026a) introduced Group Relative Policy Optimization (GRPO) Shao et al. (2024) into remote sensing. By optimizing the policy through reinforcement learning rather than SFT, it allows MLLMs to construct reasoning chains more autonomously while preserving general capabilities. Nevertheless, it still relies on human-annotated bboxes to compute accuracy rewards, such as IoU, limiting its ability to leverage large-scale unlabeled Earth observation data for training.

These limitations motivate us to seek a learning signal that remains observable without human-provided labels. One promising direction is to evaluate the semantic compatibility between a proposed region and the query. This possibility is supported by a general generation–verification asymmetry in MLLMs He et al. (2026): evaluating whether a specified candidate satisfies a requirement is often more reliable than generating a valid structured solution from a large output space. Overhead remote sensing imagery further improves the reliability of such judgments, as limited occlusion and cross-object overlap reduce semantic interference from unrelated targets. Consequently, these properties suggest that query–region compatibility can provide a viable label-free learning signal for geospatial reasoning in remote sensing.

Building on this observation, we introduce RemoteZero, a label-free framework illustrated in Fig. 1. RemoteZero employs a Generate-Crop-Verify loop, where each predicted box is converted into a visual crop and evaluated for semantic consistency with the query, providing an intrinsic reward for GRPO without human-labeled coordinates. This design enables policy optimization directly on large-scale unlabeled Earth observation data and supports two complementary learning modes: verifier-based distillation from a stronger MLLM and iterative self-evolution using the model’s own verification feedback. By leveraging the stronger verification capability of MLLMs, RemoteZero reduces label

dependence while providing a scalable route toward self-improving geospatial reasoning.

Overall, RemoteZero substantially outperforms general-purpose MLLMs and achieves a test Acc@0.5 of 71.29% on EarthReason, surpassing the strongest supervised baseline by 3.18 percentage points without using labels for policy optimization. Beyond reasoning, RemoteZero extends effectively to visual grounding and referring segmentation, achieving competitive performance against specialized supervised methods. Its performance further improves consistently across self-evolution rounds and continues to benefit from an expanding pool of unlabeled training data. Together, these results bring us one step closer to Earth observation models capable of continual self-improvement.

The contributions of this work are summarized as follows:

- We introduce RemoteZero, a label-free reinforcement learning framework that enables policy optimization directly on unlabeled Earth observation data.
- We explore a self-evolution paradigm for remote sensing MLLMs, offering a scalable route to continual improvement without additional supervision.
- We show that RemoteZero outperforms strong supervised baselines, transfers across spatial prediction tasks, and benefits from increasing training data.

2. Motivation

Our pursuit of a zero-label approach to geospatial reasoning is based on the following two observations.

2.1. Generation–Verification Asymmetry

RemoteZero is first motivated by a general generation–verification asymmetry in MLLMs [He et al. \(2026\)](#). For many structured prediction problems, generating a valid solution requires searching over a large output space, while verification only evaluates a specified candidate.

Let x denote an input and $z \in \mathcal{Z}$ denote a candidate solution. A generation model must produce a solution with high task utility:

$$z^* \in \arg \max_{z \in \mathcal{Z}} U(x, z),$$

where $U(x, z)$ measures whether z satisfies the task requirement. This formulation requires both evaluating candidate quality and searching over the output space $\mathcal{Z}$. For MLLMs, the difficulty is further amplified by autoregressive decoding: a structured output must be produced through a sequence of token decisions, and errors in intermediate tokens can invalidate the final solution.

Verification has a more constrained form:

$$V(x, z) \rightarrow s \approx U(x, z),$$

where the candidate z is already specified and the model only estimates its compatibility with the input. Thus, verification removes the open-ended search component and reduces the output interface to a scalar or binary judgment.

To examine whether this asymmetry holds in geospatial reasoning, we evaluate Qwen3-VL-8B on the EarthReason validation set under two settings. Directly generating a target region from an image–query pair yields only 43.88% Acc@0.5. When given the ground-truth region and asked to

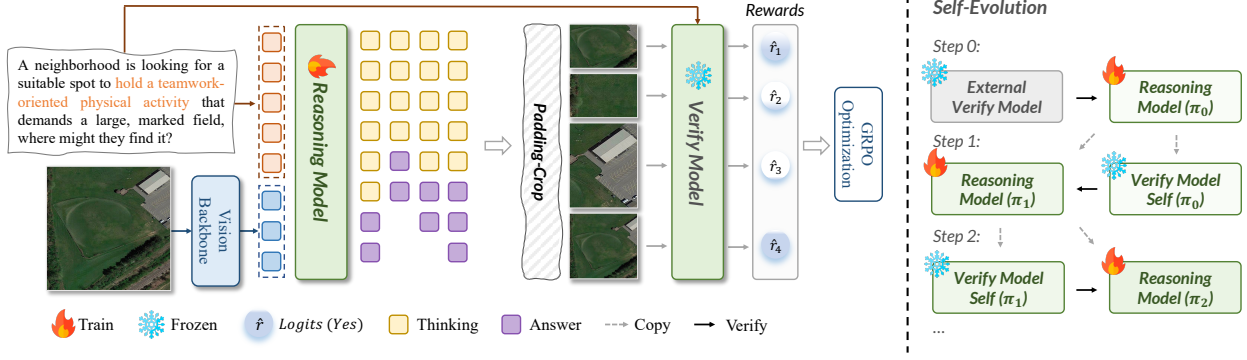


Figure 2: Overview of RemoteZero. The model generates a reasoning chain and a candidate box, which is converted into a padded crop and scored by a verifier for semantic consistency with the query. This score, combined with an area penalty, serves as the intrinsic reward for GRPO without ground-truth coordinates. RemoteZero further enables iterative self-evolution by reusing the frozen policy from the previous round as the verifier for the next round.

judge its semantic consistency with the query, the model achieves 66.73% verification accuracy. This contrast provides direct empirical evidence that region verification is substantially more reliable than coordinate generation.

2.2. Overhead Imaging Prior in Remote Sensing

Remote sensing imagery also provides a favorable domain for region-level verification. Most Earth observation images are captured from overhead or near-nadir viewpoints, where objects and land-cover areas are projected onto a nearly two-dimensional ground plane. Compared with perspective images, this view substantially reduces depth ambiguity, severe occlusion, and cross-object overlap.

Therefore, a cropped candidate region is less likely to contain multiple heavily overlapping objects from different depths. With moderate contextual padding, the crop is usually dominated by the intended facility, object, or functional area, while still preserving useful local cues such as nearby roads, buildings, fields, or open space. Thus, the semantic evidence inside a candidate crop can often be attributed to the proposed region itself rather than to unrelated overlapping content. These properties make candidate-region verification particularly suitable for geospatial reasoning.

3. RemoteZero

3.1. Problem Formulation

Given a remote sensing image $\mathcal{I}$ and a query $\mathcal{Q}$, the policy model π_θ generates a reasoning chain c and a spatial prediction b :

$$(c, b) \sim \pi_\theta(\cdot \mid \mathcal{I}, \mathcal{Q}). \quad (1)$$

The goal is to learn a policy that predicts regions satisfying the query. We formulate this objective through a reward function $\mathcal{R}(b; \mathcal{I}, \mathcal{Q})$ defined over the predicted region.

Existing GRPO-based reinforcement learning methods compute the reward by comparing the predicted region b with a human-annotated target b_{gt} :

$$\mathcal{R}_{ext}(b) = S_{gt}(b, b_{gt}), \quad (2)$$

where $S_{\text{gt}}(\cdot)$ denotes a spatial matching score based on ground-truth annotations. Therefore, the reward can only be computed when annotated spatial labels are available.

RemoteZero replaces this annotation-dependent reward with semantic verification. Given the visual region $\mathcal{I}_b$ corresponding to the prediction b , the reward is defined as:

$$\mathcal{R}_{\text{int}}(b) = V(\mathcal{I}_b, \mathcal{Q}), \quad (3)$$

where $V(\cdot)$ outputs a scalar score measuring whether the predicted region satisfies the query.

This reformulation changes the reward source from ground-truth spatial matching to query-region semantic verification, enabling policy learning without labels.

3.2. Training

Follow the previous work Yao et al. (2026a,b), we utilize GRPO to fine-tune the policy model. As shown in Fig. 2, Given a remote sensing image $\mathcal{I}$ and an implicit user query $\mathcal{Q}$, the policy model π_θ (an MLLM) generates a sequence consisting of a reasoning chain c and a predicted bounding box $\mathbf{b} = [x_1, y_1, x_2, y_2]$. This process is formulated as sampling from the policy:

$$c, \mathbf{b} \sim \pi_\theta(\cdot | \mathcal{I}, \mathcal{Q}) \quad (4)$$

Unlike standard supervised approaches, this generation process is not constrained by ground-truth coordinates but is guided solely by the subsequent reward signal.

To evaluate the correctness of the predicted location $\mathbf{b}$, we isolate the region of interest. We define a deterministic cropping function $\mathcal{T}(\cdot)$ that extracts the image patch corresponding to $\mathbf{b}$. To preserve local context essential for verification (e.g., surrounding roads or terrain), we apply a relaxed margin ratio α to the bounding box before cropping:

$$\mathcal{I}_{\text{crop}} = \mathcal{T}(\mathcal{I}, \mathbf{b}, \alpha). \quad (5)$$

The cropped patch $\mathcal{I}_{\text{crop}}$ is fed into a verifier model V . The verifier assesses the semantic entailment between the visual crop and the original query $\mathcal{Q}$. It outputs a scalar confidence score $s \in [0, 1]$, representing the probability that the cropped region semantically satisfies the query condition:

$$s = V(\mathcal{I}_{\text{crop}}, \mathcal{Q}). \quad (6)$$

It is worth noting that this formulation is agnostic to the specific instantiation of V . Whether V is an external superior model or the policy π_θ itself, the mathematical interface remains unified.

The final reward r driving the GRPO optimization is a composite of the verification confidence and a regularization term. Since the verifier might trivially award high scores to overly large crops (which are more likely to contain the target but lack precision), we introduce an area penalty:

$$r(\mathbf{b}, \mathcal{Q}) = s - \lambda \cdot \max\left(0, \frac{\text{Area}(\mathbf{b})}{\text{Area}(\mathcal{I})} - \tau\right), \quad (7)$$

where λ controls the penalty strength and τ is a threshold for acceptable area proportion. This reward is then used to compute the advantages for the group of sampled outputs in the GRPO objective.

3.3. Self-Evolution

The training procedure above can use a stronger model as the verifier V , but such an external verifier is not essential. Since verification is more reliable than generation in this setting, a model’s own verification capability can also provide useful feedback for improving its future predictions. This observation motivates a self-evolution procedure in which the policy obtained from one round serves as the verifier for the next round.

We first obtain an initial policy $\pi_\theta^{(0)}$ using an external verifier under the training procedure described in Sec. 3.2. Self-evolution then proceeds for K rounds. At round $k > 0$, a trainable policy $\pi_\theta^{(k)}$ is initialized from $\pi_\theta^{(k-1)}$, while a separate copy of the previous policy is frozen as the verifier for the current round:

$$V^{(k)}(\cdot) \leftarrow \pi_\theta^{(k-1)}(\cdot). \quad (8)$$

For each candidate box b generated by $\pi_\theta^{(k)}$, the round-specific reward follows the same crop-based verification and area regularization introduced in Sec. 3.2:

$$r^{(k)}(b, Q) = V^{(k)}(\mathcal{T}(I, b, \alpha), Q) - \lambda \max\left(0, \frac{\text{Area}(b)}{\text{Area}(I)} - \tau\right). \quad (9)$$

During optimization, $V^{(k)}$ remains fixed, while $\pi_\theta^{(k)}$ is updated with GRPO. Once the round is completed, the updated policy is carried forward and frozen as the verifier for round $k + 1$. Therefore, the external verifier is needed only to initialize the process. Subsequent rounds are driven by the preceding policy’s verification feedback and introduce no coordinate annotations.

4. Experimental Results

4.1. Experimental Setup

4.1.1. Experimental settings

We instantiate RemoteZero with Qwen3-VL-8B-Instruct [Bai et al. \(2025a\)](#) and train it with GRPO [Shao et al. \(2024\)](#) using LoRA fine-tuning [Hu et al. \(2022\)](#). Training is conducted on $8 \times$ GPUs with DeepSpeed ZeRO-2 [Rasley et al. \(2020\)](#) in bfloat16. The model is optimized for 10 epochs with a learning rate of 5×10^{-6} , using a per-device batch size of 6 and gradient accumulation of 8 steps. For each prompt, GRPO samples 4 generations with temperature 0.9. The maximum sequence length is set to 2048, and the maximum image size is capped at 802,816 pixels. Unless otherwise stated, all experiments use the same training configuration.

4.1.2. Datasets and Baselines

We conduct the geospatial reasoning experiments on EarthReason [Li et al. \(2025\)](#). To evaluate the applicability of RemoteZero to other EO tasks, we use DIOR-RSVG [Zhan et al. \(2023\)](#) for grounding and RRSIS-D [Yuan et al. \(2024\)](#) for referring segmentation. We select Qwen2.5-VL-7B [Bai et al. \(2025b\)](#), DeepSeek-VL2 [Wu et al. \(2024\)](#), InternVL3.5 [Wang et al. \(2025c\)](#), VLM-R1 [Shen et al. \(2025\)](#), GeoChat [Kuckreja et al. \(2024\)](#), and RemoteReasoner [Yao et al. \(2026a\)](#) as baselines. For visual grounding, the baselines include SkyEyeGPT [Zhan et al. \(2025\)](#), GeoChat [Kuckreja et al.](#)

Table 1: Comparison with other MLLMs on EarthReason. “*” denotes supervised by an external verifier.

Method	Acc@0.5		gIoU	
	Val	Test	Val	Test
Qwen2.5-VL-7B	41.21	45.82	38.77	41.80
Qwen3-VL-8B	43.88	48.10	41.03	44.69
DeepSeek-VL2	12.08	12.67	17.51	18.62
InternVL3.5	4.88	5.26	5.83	6.52
VLM-R1	34.64	33.31	29.67	29.44
GeoChat	10.10	8.89	12.57	11.44
RemoteReasoner	<u>66.51</u>	<u>68.11</u>	67.04	69.29
RemoteZero*	65.38	65.05	58.10	57.95
RemoteZero	69.96	71.29	<u>61.54</u>	<u>61.70</u>

Table 2: Ablation study of reward functions.

Verify	Area	Acc@0.5	
		Val	Test
✓		65.20	65.88
✓	✓	69.96	71.29

(2024), LHRS-Bot [Muhtar et al. \(2024\)](#), RemoteSAM [Yao et al. \(2025\)](#), EarthDial [Soni et al. \(2025\)](#), EarthGPT [Zhang et al. \(2024c\)](#), and GeoZero [Wang et al. \(2025a\)](#). For referring segmentation, we compare with LISA [Lai et al. \(2024\)](#), PixelLM [Ren et al. \(2024\)](#), Text4Seg++ [Lan et al. \(2025\)](#), GeoGround [Zhou et al. \(2024a\)](#), SegEarth-R1 [Li et al. \(2025\)](#), and ProVG [Li et al. \(2026a\)](#).

4.2. Main Results

Table 1 compares RemoteZero with general-purpose MLLMs, remote-sensing MLLMs, and supervised geospatial reasoning methods on EarthReason. Existing MLLMs exhibit limited performance on implicit geospatial localization. Among the general-purpose models, Qwen3-VL-8B achieves the strongest test Acc@0.5 of 48.10%, while reasoning-oriented VLM-R1 reaches only 33.31%. GeoChat also performs poorly despite its remote-sensing specialization, suggesting that domain-specific visual-language alignment alone is insufficient for resolving complex user intent into precise regions.

RemoteZero substantially improves over these baselines. With an external verifier, RemoteZero* achieves 65.05% test Acc@0.5 without using human-provided solution labels for policy optimization, approaching the fully supervised RemoteReasoner. After self-evolution, RemoteZero further improves to 71.29%, surpassing RemoteReasoner by 3.18 percentage points and achieving the best Acc@0.5 on both splits. Nevertheless, RemoteZero obtains a lower test gIoU than RemoteReasoner, which indicates that the current verifier reward is more effective at identifying semantically correct regions than at calibrating precise spatial extents. We view this as an important direction for future improvement.

Table 3: Ablation study of cropping strategies with different context padding ratios.

Crop Strategy	Padding	Acc@0.5	
		Val	Test
Strict	–	64.61	65.13
Context	10%	66.22	68.84
	15%	69.96	71.29
	20%	67.06	68.17

Table 4: Ablation study of different verifier models.

Verify Model	Acc@0.5		Verify Score
	Val	Test	
Qwen2.5-VL-72B	65.38	65.05	66.19
Qwen2.5-VL-32B	63.01	62.85	63.25
Qwen3-VL-32B	64.55	64.90	67.30
Qwen3-VL-8B	63.52	64.21	<u>66.73</u>
RemoteReasoner	63.68	64.17	65.48

4.3. Ablation Studies

4.3.1. Ablation on Reward Design

To assess whether semantic verification alone is sufficient for spatially precise reward learning, We compare the verifier-only reward with the full reward that further includes area penalty, as reported in Table 2. Using only the verification score already provides a meaningful training signal, achieving 65.20% validation and 65.88% test Acc@0.5. However, this reward can favor overly large boxes, since larger regions are more likely to contain query-relevant content and receive positive verifier feedback. Adding the area penalty improves performance to 69.96% on validation and 71.29% on test, indicating that a lightweight geometric constraint is necessary to prevent trivial large-region predictions. These results suggest that semantic verification provides the core learning signal, while area penalty further improves localization precision.

4.3.2. Ablation on Cropping Strategy

We compare strict cropping with context crops using different padding ratios to examine how the amount of visual context affects verification-based training. As illustrated in Table 3, strict cropping achieves 64.61% validation and 65.13% test Acc@0.5, suggesting that the predicted box alone may discard surrounding cues needed for interpreting functional geospatial intent. Introducing contextual padding improves performance, with 15% padding reaching the best results of 69.96% on validation and 71.29% on test. However, further increasing the padding to 20% leads to lower accuracy, indicating that excessive context may introduce irrelevant regions and weaken the verifier’s region-specific judgment.

4.3.3. Ablation on Verifier Models

To study the effect of verifier choice, we train the same base model using rewards provided by different verifier models, as shown in Table 4. The verification score measures the average probability

Table 5: Analysis of verification reward behavior under different visual inputs.

Input Type	Prob. (Yes) %	Rewards	Acc@0.5
GT box	86.3	0.770	65.13
Shifted box	34.9	0.302	20.08
Padding image	93.5	0.896	71.29
Whole image	11.7	0.091	16.53
No targets	9.0	0.074	15.40

of answering Yes when the verifier is given a ground-truth crop together with its query, while Val and Test report the downstream Acc@0.5 of the trained base model. Within the same model family, larger verifiers generally achieve higher verification scores, and the newer Qwen3-VL models also exhibit stronger verification capability. However, higher verification scores do not always lead to proportional gains in downstream localization accuracy: Qwen2.5-VL-72B achieves the best Val and Test results, whereas Qwen3-VL-32B obtains the highest verification score. Considering the substantial training cost of larger verifiers and the relatively small difference in final accuracy, we adopt Qwen3-VL-8B as the default verifier, which provides a favorable trade-off between verification quality and training efficiency.

4.4. Further Analysis

4.4.1. Is the Verification Reward Vulnerable to Trivial Coverage Bias?

To examine whether the verification reward reflects genuine query-region compatibility, we evaluate the verifier under different visual inputs, as reported in Table 5. The verifier assigns high confidence to ground-truth regions, while shifted boxes and target-free regions receive substantially lower rewards, indicating that the reward is sensitive to spatial relevance rather than arbitrary visual content. The padded crop achieves the highest verification confidence and final accuracy, suggesting that moderate local context helps the verifier interpret functional geospatial intent. In contrast, the whole image receives a much lower reward despite containing the target and more visual content, showing that the verifier does not simply favor larger regions. These results indicate that the verification reward primarily captures query-region compatibility, while padded crop provides an effective balance between semantic context and spatial specificity.

4.4.2. Can RemoteZero Be Applied to Other Tasks?

To examine whether RemoteZero can be applied to other remote sensing tasks, we instantiate the proposed label-free training framework on visual grounding and further extend it to referring segmentation, as reported in Table 6. For visual grounding, RemoteZero achieves 76.0 Pr@0.5 and 68.9 mIoU, outperforming most specialized grounding methods and remaining close to the strongest supervised baseline. For referring segmentation, we use the bounding boxes predicted by the label-free RemoteZero grounding model as prompts for SAM 2, without introducing segmentation labels or segmentation-specific training for RemoteZero. This simple combination achieves 75.3 Pr@0.5 and 65.4 mIoU, reaching performance comparable to dedicated referring segmentation methods. These results show that RemoteZero is not tied to a particular geospatial reasoning task: its verification-based reward formulation can be instantiated for other spatial prediction tasks and can also provide effective intermediate outputs for downstream pixel-level applications.

Table 6: Application of RemoteZero to visual grounding and referring segmentation.

Task	Method	$Pr@0.5$	$mIoU$
Visual Grounding	SkyEyeGPT	70.5	–
	GeoChat	62.0	–
	LHRS-Bot	73.5	–
	RemoteSAM	74.4	65.1
	EarthDial	46.1	39.5
	EarthGPT	76.7	69.3
	RSGround-R1	71.8	63.4
	TinyRS-R1	74.9	–
	GeoZero	75.7	–
	RemoteZero	<u>76.0</u>	<u>68.9</u>
Referring Segmentation	LISA	–	26.8
	PixelLM	–	31.7
	Tex4Seg++	73.2	62.8
	GeoGround	–	60.5
	SegEarth-R1	67.6	66.4
	ProVG	76.1	<u>65.4</u>
	RemoteZero [†]	<u>75.3</u>	<u>65.4</u>

4.4.3. How Many Self-Evolution Rounds Are Needed?

We also evaluate RemoteZero after each evolution round on EarthReason to study how iterative self-evolution affects performance. As shown in Fig. 3, both validation and test accuracy improve consistently across the six rounds. From the first to the sixth round, validation Acc@0.5 increases from 61.70% to 69.96%, while test accuracy improves from 63.19% to 71.29%, corresponding to gains of 8.26 and 8.10 percentage points, respectively. The largest improvements occur during the early rounds, whereas the gains from the fifth to the sixth round decrease to 0.18 and 0.52 percentage points on the validation and test sets. This trend suggests that self-evolution is approaching saturation by the sixth round. We therefore use the 6th-round model for the final evaluation, providing a practical balance between performance and training cost.

4.4.4. Can Self-Evolution Continue with Expanding Data?

The previous experiment shows that self-evolution gradually saturates after six rounds when the training data remain fixed. To examine whether this plateau results from limited data scale, we simulate a continually expanding data pool by sampling queries from EarthReason and doubling the accumulated training data at each version. As shown in Fig. 4, unlike the saturation observed with a fixed training set, both metrics retain substantial gains in later versions. These results suggest that the earlier plateau is primarily associated with the finite data pool rather than an inherent limit of self-evolution. As increasingly diverse queries become available, the verifier becomes more reliable and provides stronger feedback for subsequent policy optimization.

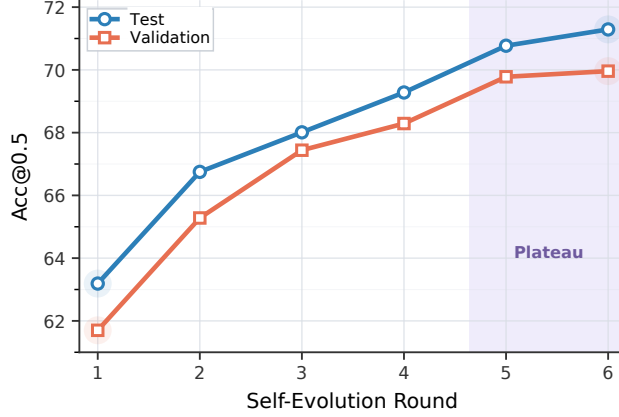


Figure 3: Performance across six self-evolution rounds on EarthReason. Both validation and test Acc@0.5 improve consistently, while the gains diminish in later rounds, indicating saturation under a fixed training set.

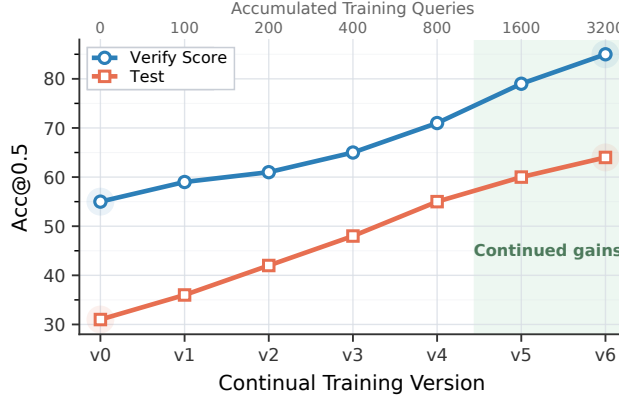


Figure 4: Performance under continual data scaling on EarthReason. The accumulated training queries double at each version. Both the verifier’s Verify Score and the policy’s test Acc@0.5 continue to improve, suggesting that the earlier plateau is mainly associated with limited data diversity.

4.4.5. Qualitative Case Study

Fig. 5 presents representative successful and failed cases of RemoteZero. In the first row, the model correctly interprets the emergency evacuation intent, identifies the marked helipad as a flat and unobstructed landing area, and produces a compact bounding box around it. The verifier confirms that the predicted region satisfies the query, yielding a high reward of 0.988. In the second row, however, the model overemphasizes the broad and open characteristics of the sandy beach while overlooking the query’s requirement for human-made facilities that support efficient goods handling. Consequently, it incorrectly reasons that the beach is the target and predicts an excessively large region instead of focusing on the relevant coastal infrastructure. The verifier successfully detects this semantic mismatch, outputs *No*, and assigns zero reward. This failure case demonstrates that verification-based feedback can reject superficially plausible predictions when their localized regions do not satisfy all essential constraints expressed in the query.

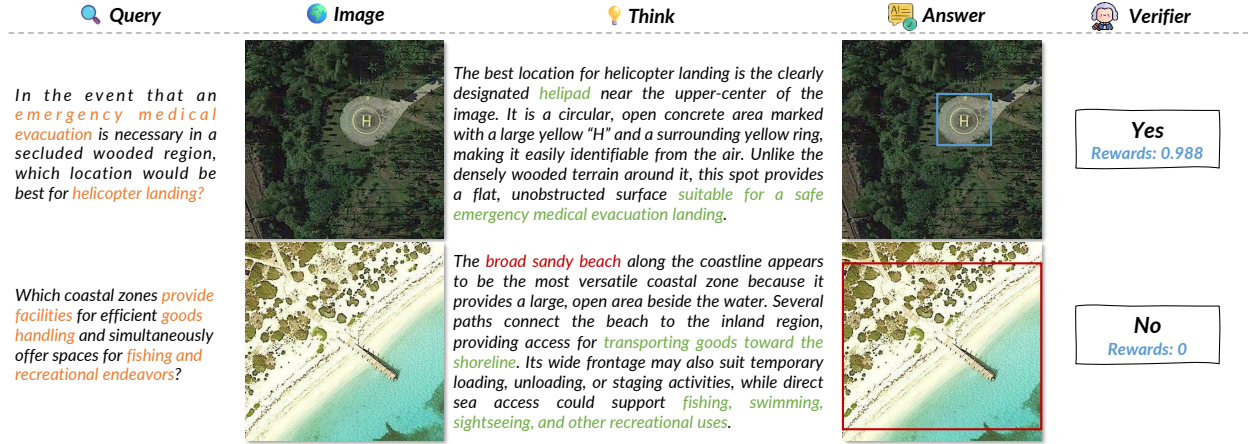


Figure 5: Qualitative case studies. The verifier rewards accurate localization (top) while rejecting a semantically mismatched, overly large prediction (bottom).

5. Related Work

5.1. Remote Sensing Multi-modal Models

The adaptation of MLLMs to remote sensing [Ma et al. \(2025, 2026\)](#), [Min et al. \(2026\)](#), [Ye et al. \(2026a,b\)](#) began with domain-specific captioning and grounded dialogue, represented by RSGPT [Hu et al. \(2025\)](#) and GeoChat [Kuckreja et al. \(2024\)](#). EarthGPT [Zhang et al. \(2024c\)](#) and SkyEyeGPT [Zhan et al. \(2025\)](#) then expanded toward unified multi-sensor interpretation and instruction tuning. Subsequent works improved granularity and interaction through relational reasoning in EarthVQA [Wang et al. \(2024a\)](#), VGI-enhanced training in LHRS-Bot [Muhtar et al. \(2024\)](#), fine-grained instruction tuning in SkySenseGPT [Luo et al. \(2024\)](#), and visual prompting in EarthMarker [Zhang et al. \(2024b\)](#). TEOChat [Irvin et al. \(2025\)](#) and RingMoGPT [Wang et al. \(2024b\)](#) further extended temporal and grounded understanding, while RSUniVLM [Liu and Lian \(2024\)](#), Falcon [kelu et al. \(2025\)](#), and EagleVision [Jiang et al. \(2025\)](#) moved toward unified pixel-level and object-centric interpretation.

5.2. Geospatial Reasoning Models

Remote sensing research has recently moved beyond standard perception [Liu et al. \(2024\)](#), [Huang et al. \(2026\)](#) toward complex geospatial reasoning with MLLMs. SegEarth-R1/2 [Li et al. \(2025\)](#), [Xin et al. \(2026\)](#) addressed implicit queries through pixel-level reasoning, followed by RemoteReasoner [Yao et al. \(2026a\)](#), which introduced a unified RL workflow for autonomous multi-granularity analysis. Subsequent work increasingly adopted RFT and CoT: Geo-R1 [Zhang et al. \(2026\)](#) applied RFT to few-shot referring expression understanding, GeoVLM-R1 [Fiaz et al. \(2025\)](#) designed task-aware rewards for diverse observation tasks, and RSThinker [Liu et al. \(2025\)](#) developed perceptually grounded CoT to produce verifiable outputs. More recently, GeoZero [Wang et al. \(2025a\)](#) encouraged emergent reasoning without predefined CoT supervision through answer-anchored policy optimization, while GeoReason [Li et al. \(2026b\)](#) used consistency-aware RL to better align internal reasoning with final predictions.

6. Conclusion

We presented RemoteZero, a label-free framework for reinforcement-based geospatial reasoning. RemoteZero replaces human-provided solution labels with intrinsic semantic verification and further supports self-evolution by reusing previous-round models as verifiers. Experiments on EarthReason show that RemoteZero achieves 71.29% Acc@0.5, surpassing the strongest supervised baseline by 3.18 percentage points. We are excited by the broader possibility that our method may offer a general route for MLLMs to learn from unlabeled data and gain continually improvement.

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